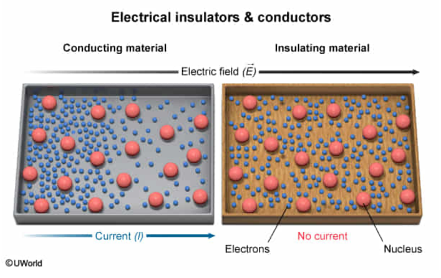
The misdirected transmission of a high-voltage electrical current is associated with a risk of accidental fire. To eliminate the risk of such accidents, the type of material that must be applied to the surface of bare wires is known as:

- A. an electrical conductor. [2%]
- ✔ B. an electrical insulator. [81%]
- C. a thermal conductor. [1%]
- D. a thermal insulator. [15%]

Correct
81% answered correctly

Explanation:



Electrical conductivity is a physical property that describes how easily electric charge flows through a given material. In solids, electrical conductivity is closely related to how readily electrons can move between atoms to produce an **electric current**.

Electrons within metals are weakly attracted to their corresponding nuclei because the atomic number of many metals is such that the distance between positively charged protons and negatively charged valence electrons is relatively great. Consequently, electrons within metallic materials may be easily dislodged following exposure to an external electric field. Therefore, most metals are **electrical conductors** whereas most nonmetals are **electrical insulators** (materials that do not readily transmit electrical energy).

In many cases, electrical fires begin as combustible foreign materials enter the electrical circuit, accumulate thermal energy from the high voltage source, and ignite. To prevent such accidental fires, electrical wires are coated in with an electrical insulator with high resistance, which does not transmit electrical energy and prevents electric current from easily exiting the circuit.

(Choice A) Electrical conductors are used to facilitate current because the electric field that must be applied to a highly conductive material is *less* than that required for other materials, which conserves electrical potential energy (ie, voltage).

(Choice C) Many metals are also **thermal conductors**, meaning that they readily act to transfer thermal (heat) energy via conduction. Facilitating thermal energy transfers between nearby atoms and molecules will not prevent adverse electrical events like electrical fires.

(Choice D) Many thermal insulators are materials in which atoms and molecules are relatively sparse (eg, foam),

making the exchange of thermal energy through conduction difficult. Thermal insulation is used to maintain the temperature of a region or object, not to prevent electrical accidents.

Educational objective:
Electrical conductors facilitate electrical current (ie, the movement of charge) whereas electrical insulators inhibit current. Although some electrical conductors are also thermal conductors, mechanisms of thermal and electrical conductivity are not the same.

Time spent:0 sec
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